

Anish Rao Toorpu

AI / ML Engineer

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PROFESSIONAL SUMMARY

- Around nine years of experience as an AI and machine learning engineer, building and deploying models into production, with the last three years dedicated to Generative AI and large language model applications across banking, healthcare and insurance.
- Strong engineering mindset, treating machine learning work as software that must be tested, deployed, monitored and maintained, not just as experiments that stay inside notebooks.
- Designed and shipped Retrieval Augmented Generation systems end to end, covering document ingestion, chunking, embeddings, vector storage with FAISS and Pinecone, retrieval, reranking and response generation.
- Built LLM powered applications using GPT-4o, Claude, Llama 3 and Mistral, handling prompt engineering, function calling, structured outputs, streaming responses and careful evaluation of quality and cost.
- Implemented Model Context Protocol servers and clients so that LLM agents connect to internal tools, APIs and data sources through a standard protocol, which makes integrations cleaner and easier to maintain.
- Built agentic workflows where LLMs can plan steps, call tools and work through multi step tasks, with sensible limits, logging and fallbacks, so behavior stays predictable.
- Experienced in fine tuning open-source models with parameter efficient methods such as LoRA and QLoRA, and in applying quantization so models fit within available GPU and cost budgets.
- Comfortable across the machine learning lifecycle, from feature engineering and model training with scikit-learn, XGBoost and PyTorch, through to packaging models as services and deploying them to the cloud.
- Solid hands-on MLOps experience, including building CI/CD for model deployment, containerizing services with Docker, tracking experiments with MLflow, and setting up monitoring for drift and performance.
- Strong Python engineer who also writes SQL and PySpark, builds clean APIs with FastAPI, and cares about readable, well-structured and well tested code.
- Used to working in cross functional teams with data scientists, backend engineers, product owners and platform teams, and comfortable owning a feature from design through deployment and support.
- Reliable and thorough, with a habit of writing clear documentation and design notes so that systems can be understood, handed over and extended by the rest of the team.

TECHNICAL SKILLS

Generative AI & LLMs	GPT-4o, Claude, Azure OpenAI, RAG, LangChain, LlamaIndex, Hugging Face, Prompt engineering, Agentic workflows, Model Context Protocol (MCP), LangGraph
LLM Engineering	Embeddings, Chunking, Reranking, Context management, Evaluation, Guardrails, Prompt versioning, Token and cost optimization, Streaming, Caching
Model Tuning	LoRA, QLoRA, Parameter efficient fine tuning, quantization, Transformer fine tuning, BERT, Model distillation basics
Vector Databases	FAISS, Pinecone, ChromaDB, Weaviate, OpenSearch, Aurora PostgreSQL
Machine Learning	Scikit-learn, XGBoost, LightGBM, PyTorch, classification, regression, clustering, feature engineering, model evaluation
Programming	Python, SQL, PySpark, Polars, FastAPI, Flask, REST APIs, Pandas, NumPy
MLOps & Cloud	AWS (SageMaker, EC2, S3, Lambda), Azure ML, GCP (Vertex AI, Cloud Storage, BigQuery) Docker, Kubernetes basics, MLflow, GitHub Actions, CI/CD, Airflow, model monitoring
Tools & Practices	Git, Jira, Confluence, Agile/Scrum, unit testing, code review, technical documentation

PROFESSIONAL EXPERIENCE

BNY Mellon

Jan 2025 – Present

AI / ML Engineer – Generative AI

New York, NY

Project Overview: Worked on building and deploying Generative AI services for a large US custody and asset servicing bank, covering Retrieval Augmented Generation, agentic workflows and Model Context Protocol integrations. Worked on shipping these as production services with APIs, containers and CI/CD, deployed and monitored on AWS.

Roles & Responsibilities:

- Worked on building and deploying Retrieval Augmented Generation services that let internal teams query large collections of policy, procedure and product documents and get grounded answers with citations.
- Built document ingestion, parsing, chunking, and embedding pipelines for PDFs, SharePoint content, and enterprise APIs, improving retrieval quality and grounding accuracy for RAG systems.
- Built multi-step agentic workflows using LangGraph for orchestration, tool calling, fallback handling, and controlled execution flows in enterprise GenAI systems.
- Implemented Model Context Protocol servers that expose internal document stores, databases and tools to LLM agents through a standard interface, which cut down repeated custom integration work.
- Developed agentic workflows where the LLM can call MCP tools to look up information and complete multi step tasks, with step limits, logging and fallback handling built in.
- Built and exposed GenAI services as FastAPI endpoints, containerized them with Docker, and worked with the platform team on deployment, scaling and observability.
- Deployed GenAI services on AWS, using SageMaker for hosted model endpoints, S3 for document and embedding storage, Lambda for processing steps, and CloudWatch for logging and monitoring.
- Worked on AWS based scaling, secure access to internal data sources, and cost monitoring so GenAI services stayed reliable and predictable in cost as adoption grew.
- Set up CI/CD pipelines for GenAI services so that changes to prompts, retrieval logic and code went through automated testing before release.
- Evaluated cloud-native deployment patterns across AWS and GCP environments for scalable GenAI services and retrieval workflows.
- Created an LLM evaluation harness combining automated scoring and human review to check faithfulness, relevance and safety before features shipped.
- Added guardrails, input validation and prompt safety checks, and worked with risk partners to define what acceptable model behavior looked like.
- Compared GPT-4o, Claude and open weight models on accuracy, latency and cost, and implemented model routing, so each use case used the most suitable option.
- Tracked token usage, latency and cost per service, and improved performance through caching, retrieval tuning and prompt optimization.
- Maintained clear documentation of architecture, MCP integrations and deployment steps in Confluence, so the platform stayed maintainable as it grew.

Environment: *Python, GPT-4o, Claude, Llama 3, LangChain, LangGraph, LlamaIndex, Model Context Protocol (MCP), FAISS, Pinecone, Hugging Face, FastAPI, Docker, MLflow, AWS, GitHub Actions, Git, Jira*

Cigna

Sep 2023 – Dec 2024

Machine Learning Engineer

Bloomfield, CT

Project Overview: Worked on building and deploying NLP and machine learning models for a US health services company, processing clinical notes and transcripts into structured data for claims and clinical operations. Worked on transformer-based models, model APIs, and deployment of services on Azure ML.

Roles & Responsibilities:

- Worked on building and deploying NLP models that processed clinical notes and call transcripts to extract structured information for operations and analytics teams.
- Built named entity recognition and text classification services using spaCy and Hugging Face transformers and packaged them as APIs for other teams to use.
- Evaluated and integrated Azure OpenAI based workflows for enterprise NLP and Retrieval-Augmented Generation use cases on healthcare policy and clinical datasets.
- Developed an early Retrieval Augmented Generation prototype on policy and coverage documents, which gave the team practical experience that fed into later GenAI work.
- Fine-tuned transformer models on healthcare text using parameter efficient methods, and handled data preparation, evaluation and review with subject matter experts.
- Built machine learning models for claim triage and prioritization using gradient boosting, and worked on feature engineering over large, structured datasets.
- Built PySpark-based ETL pipelines for large healthcare datasets and document processing workflows supporting downstream ML and retrieval systems.
- Containerized models with Docker and worked with engineering teams to deploy them into existing case management workflows.
- Deployed NLP and machine learning models across Azure ML and GCP-based AI/ML environments for training, hosting, experimentation, and scalable processing of healthcare datasets.
- Used MLflow to track experiments and model versions, which kept the team's work organized and easy to compare.
- Set up monitoring for deployed models to watch for drift and performance drops and handle retraining when input data shifted.
- Worked within strict data privacy and compliance controls given the sensitivity of healthcare data.
- Collaborated with data scientists and clinical experts to validate outputs before models were used in operations.
- Documented model design, APIs and deployment steps so solutions can be supported and handed over cleanly.

Environment: *Python, spaCy, Hugging Face Transformers, BERT, scikit-learn, XGBoost, PyTorch, PySpark, SQL, RAG, LangChain, Flask, MLflow, Azure ML, Docker, Git*

Nationwide

Machine Learning Engineer

Jan 2022 – May 2023

Columbus, OH

Project Overview: Worked on building and deploying predictive models and text analytics for a large US insurance and financial services company, supporting underwriting and claims. Worked on moving models from one off scripts into repeatable scheduled pipelines, containerized with Docker and run on AWS.

Roles & Responsibilities:

- Worked on building predictive models for underwriting and claims use cases, taking them from exploration through to deployment.
- Built text analytics on adjuster notes and claim descriptions using TF-IDF and embeddings to support classification and pattern detection.
- Developed classification and regression models with scikit-learn and XGBoost, and put real effort into validation and avoiding overfitting.
- Worked with engineering teams to move models into repeatable scheduled pipelines instead of one-off scripts.
- Wrote SQL and Python to pull, clean and join data from several internal systems, and handle real world data quality problems.
- Containerized model services with Docker and supported their deployment into internal environments.

- Worked with cloud-based data and compute on AWS, using S3 for datasets and EC2 for model training, and helped move model workloads off local machines into the cloud.
- Built dashboards so business partners could see model outputs and trends clearly.
- Documented modeling assumptions, data definitions and limitations so that work can be reviewed and reused.
- Took part in model review sessions and refined approaches based on feedback from business and technical partners.
- Set up basic monitoring to track model performance over time after deployment.
- Supported A/B style checks to confirm model driven changes actually improved outcomes.
- Helped junior team members with Python and modeling questions and reviewed their work.

Environment: *Python, SQL, scikit-learn, XGBoost, Pandas, NumPy, NLP, TF-IDF, Docker, MLflow, Tableau, Git, Jupyter*

Citibank

Dec 2019 – Dec 2021

Machine Learning Engineer

New York, NY

Project Overview: Worked on building and deploying machine learning models for fraud monitoring, customer analytics, and risk reporting at a global bank. Worked on classification models over large transaction datasets, batch scoring pipelines, and processing on the bank's Hadoop based data platform.

Roles & Responsibilities:

- Worked on building machine learning models for fraud and risk detection using large transaction datasets and handled the class imbalance common to rare event problems.
- Developed customer analytics models including segmentation and churn analysis to support business teams.
- Built classification models with scikit-learn and gradient boosting, focusing on precision and recall tradeoffs that mattered for fraud detection.
- Worked with engineering teams to deploy models into batch scoring pipelines with reliable scheduling and delivery.
- Worked with the bank's Hadoop and Hive based data platform to process large transaction datasets and used internal compute environments for model training and scoring.
- Wrote complex SQL to extract and aggregate data from large banking data warehouses and built reproducible preparation steps in Python.
- Started working with customer interaction text, applying basic NLP for classification and keyword extraction.
- Documented model logic and validation to meet internal model governance expectations in a regulated environment.
- Built dashboards and reports giving risk and business teams a clear view of model outputs and metrics.
- Took part in model reviews with risk stakeholders and adjusted features and thresholds based on their input.
- Investigated data quality issues across source systems and worked with data owners to resolve them.
- Mentored newer team members on Python and good engineering practice.
- Communicated technical results clearly, so non-technical partners could act on them confidently.

Environment: *Python, SQL, scikit-learn, XGBoost, Pandas, NumPy, NLP, Tableau, Git, Jupyter, Hadoop, Hive*

LatentView Analytics

Apr 2017 – Oct 2019

Junior ML Engineer

Chennai, India

Project Overview: Worked on analytics and early machine learning projects for consumer and retail clients at a data and analytics services company. Worked on data analysis, reporting, dashboards and early predictive models, building a strong foundation in data handling and engineering practice.

Roles & Responsibilities:

- Started my career on analytics projects for consumer and retail clients, building analysis and reporting that supported business decisions.

- Wrote SQL to pull and transform data from client databases and used Python to clean and analyze it.
- Built early predictive models for use cases like customer segmentation using scikit-learn.
- Created dashboards and visualizations in Tableau so client teams could track key metrics.
- Worked with senior team members to understand client requirements and turn them into clear tasks.
- Performed exploratory data analysis to find trends and data quality issues and summarized findings.
- Learned and applied core statistics and machine learning concepts through project work.
- Took part in client calls to walk through analysis and answer questions.
- Documented analysis steps and assumptions so work could be checked and repeated.
- Built early habits around clean code, version control and structured project work.
- Supported multiple projects in parallel and learned to manage priorities.
- Built a strong foundation in data handling and analytical thinking.

Environment: *Python, SQL, scikit-learn, Pandas, NumPy, Excel, Tableau, Statistics, Jupyter*

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science.

May 2017

India